SmartPot

COMPONENT TECHNICAL DOCUMENTATION

SmartPot-API-Java (Legacy Stack A)

Legacy Spring Boot Core API and MQTT Bridge

REST with JWT, MQTT broker bridge, physical and virtual crops,
AI automation agent, notification channels, security,
configuration, and operations of the legacy Stack A API.



DOCUMENT / DOCUMENTO VERSION / VERSIÓN
SmartPot-API-Java Documentation

TEAM / EQUIPO
1.1 · October 2026

PROJECT / PROYECTO
SmartPotTech smartpot.app

> **WARNING / ATENCIÓN**
> **Legacy Component (Stack A)**: This service has been superseded in production by the canonical Go API (`SmartPot-API`). This document preserves the historical architecture and implementation details of the original Spring Boot 4.1 and Java 21 service.

DOCUMENT

Document Summary

FIELD	VALUE
Project	SmartPot · smartpot.app
Component	SmartPot-API
Version	1.1 · octubre 2026
Scope	API domains, physical and virtual crops, telemetry and command flow, security, configuration, testing, and operations
Platform Documentation	Technical Documentation, Project Journey, Software Lifecycle, and Global Diagrams
Maintenance	Generated from `docs/` in this repository via `.github/docs/tools`; updated with every component change

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8. Automated Testing Suite
9. Deployment and Production Operations

PART I

Component Overview

SECTION 01

Purpose

In Plain Terms

The API serves as the primary gateway of SmartPot. The PWA communicates via HTTPS; crop devices communicate via MQTT through the broker. The API stores telemetry readings, queries the AI assistant for agronomic health evaluations, dispatches actuation commands, provisions per-crop MQTT credentials, and delivers alerts through the PWA and Telegram. All user interactions route through this layer, scoped strictly to authenticated tenant accounts.

RESPONSIBILITY	MECHANISM
Accounts & Auth	Registration, JWT login, email password recovery, user profile, and cascaded account deletion
Crops	Physical or virtual (immutable once created), 6 species profiles, 4 system forms, live weather location, and automatic mode
Devices	Per-crop MQTT tenant account with 192-bit AES-256-GCM encrypted credentials; Mosquitto dynamic provisioning
Telemetry	MQTT telemetry validated against physical sensor bounds, historical series, digest recap, and CSV export
Commands	Actuators operated as switches (409 Conflict if redundant or pending), QoS 1 commands, device ACK, expiration ticker, and bulk orders
AI Assistant	Evaluation with SmartPot-AI-Python, automation reactive agent, and real-crop telemetry pipeline for continuous learning
Simulation	Virtual crop parameters and internal proxy to SmartPot-DataGenerator-Python
Alerts	PWA and Telegram alerts, pluggable notification channels, per-crop subscriptions, daily digests, and shared collaborator chats

SECTION 02

Component Architecture

Each domain follows a layered architecture (`controller`, `service`, `repository`, `mapper`, and `model`). Domains interact via service interfaces and internal Spring ApplicationEvents (`ReadingRecordedEvent`, `CropDeletedEvent`, `DeviceKeyRotatedEvent`, `NotificationCreatedEvent`), isolating domain concerns.

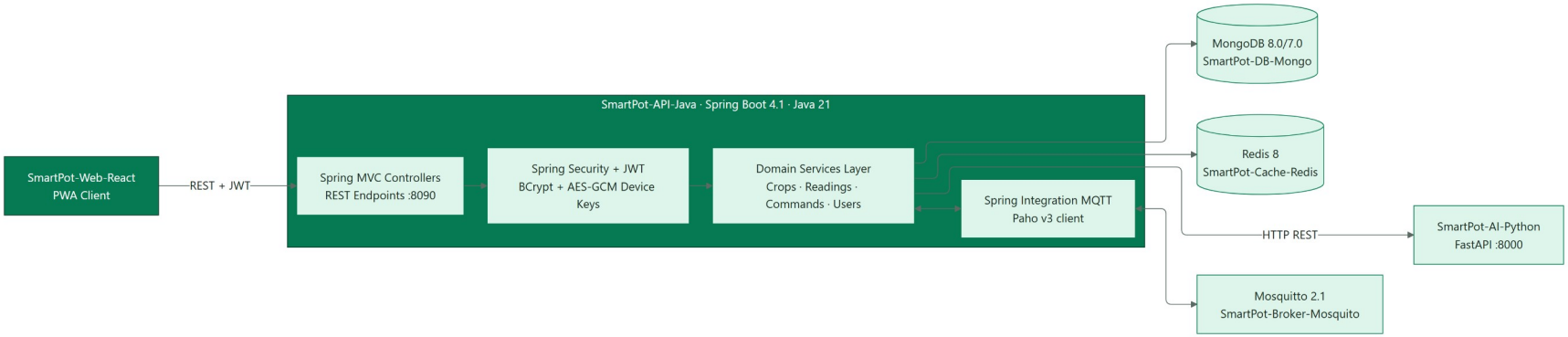


Figure / Figura 1 · *SmartPot-API Internal Architecture*

PACKAGE	DESCRIPTION
`security, users`	Security filters, JWT authentication, rate limiting, AES-GCM encryption, user management
`crops`	Crop domain, immutable kind, system form, automatic mode, and device credentials
`readings`	Telemetry ingest, physical boundary validation, historical series, CSV export
`actuators, commands`	Crop actuators state machine and command lifecycle management
`mqtt`	Paho MQTT client, v1 topic routing, telemetry decoder, dynamic broker security provisioning
`ai`	AI client, periodic evaluation, reactive automation agent, and `LearningFeed` queue
`overview`	Fleet overview: aggregates, `$dateTrunc` time-series metrics, and crop comparator
`notifications, channels`	Alert routing and external channels (`NotificationChannel`, Telegram Bot)
`virtualdevices`	Virtual crop simulation proxy, reconciliation loops, and legacy backfill
`cache, mail, health, config`	Redis cache with in-memory fallback, JavaMail, `/health` endpoint, Spring configuration

SECTION 03

Physical and Virtual Crops

When creating a crop, its origin type is selected once and remains immutable. Both physical and virtual crops have dedicated credentials on the MQTT broker; they differ in who publishes telemetry.

POLICY	PHYSICAL (REAL)	VIRTUAL
Telemetry Publisher	Physical ESP32 or simulated in Wokwi	SmartPot-DataGenerator
Credentials	Displayed once and rotatable	Internal: API provisions simulator over docker network
Initial Actuators	Water pump, UV light, ventilation fan	All six supported actuators
Dedicated Endpoints	`/device` and `/device/key`	`/virtual-device` (PUT updates, DELETE pauses)
Continuous Learning	Telemetry trains ML models	Excluded from continuous learning dataset
Tenant Quota	Max 20 crops per account	Up to 5 virtual crops per account

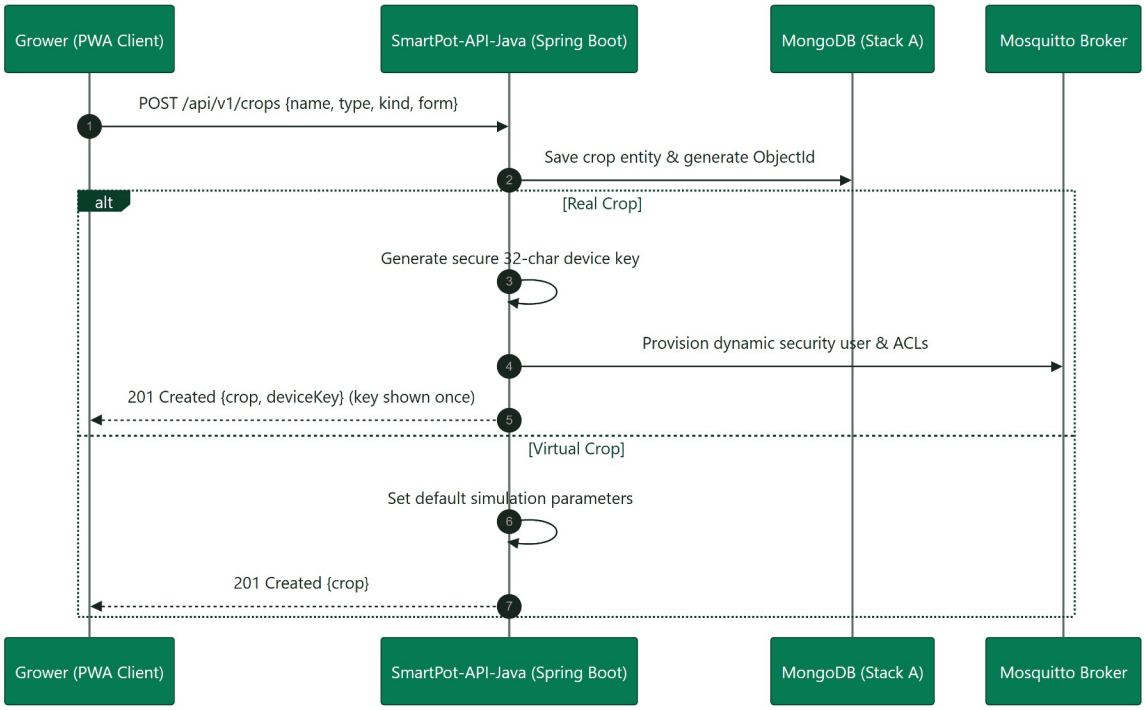


Figure / Figura 2 · *Crop Creation Flow (Physical vs. Virtual)*

The system form (`POT`, `NFT`, `TOWER`, `RAFT`) configures visual rendering in the PWA and can be edited anytime. Legacy crops created prior to explicit typing are normalized on startup via `CropKindBackfill`.

PART II

System Operations & Contracts

SECTION 04

Telemetry Ingestion to Command Execution

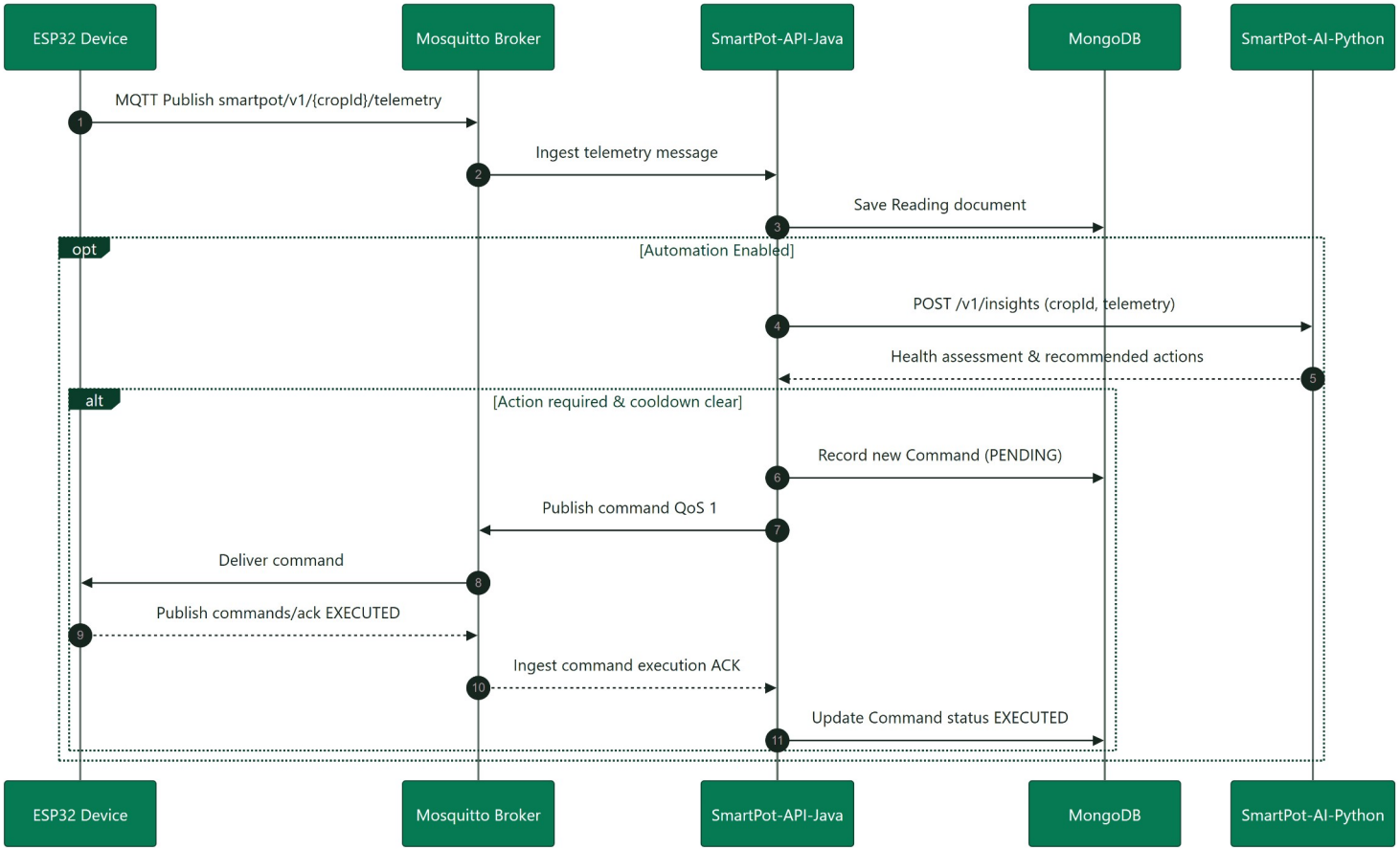


Figure / Figura 3 · *Telemetry to Command Lifecycle*

POLICY	VALUE
Crop Telemetry Rate Limit	1 reading per 5 seconds max; bursts discarded
Sensor Validation	Every metric within physical sensor bounds; invalid samples discarded
AI Evaluation Cadence	Max once per 30s in automatic mode, 5min in manual mode
Agent Actuator Cooldown	10 minutes per actuator
Command Expiration	2 minutes without ACK → `EXPIRED` status and alert
Learning Ingest Batch	Flushed every 60s, up to 1,000 readings per batch; 20,000 capacity

SECTION 05

Request Security & Authorization Pipeline

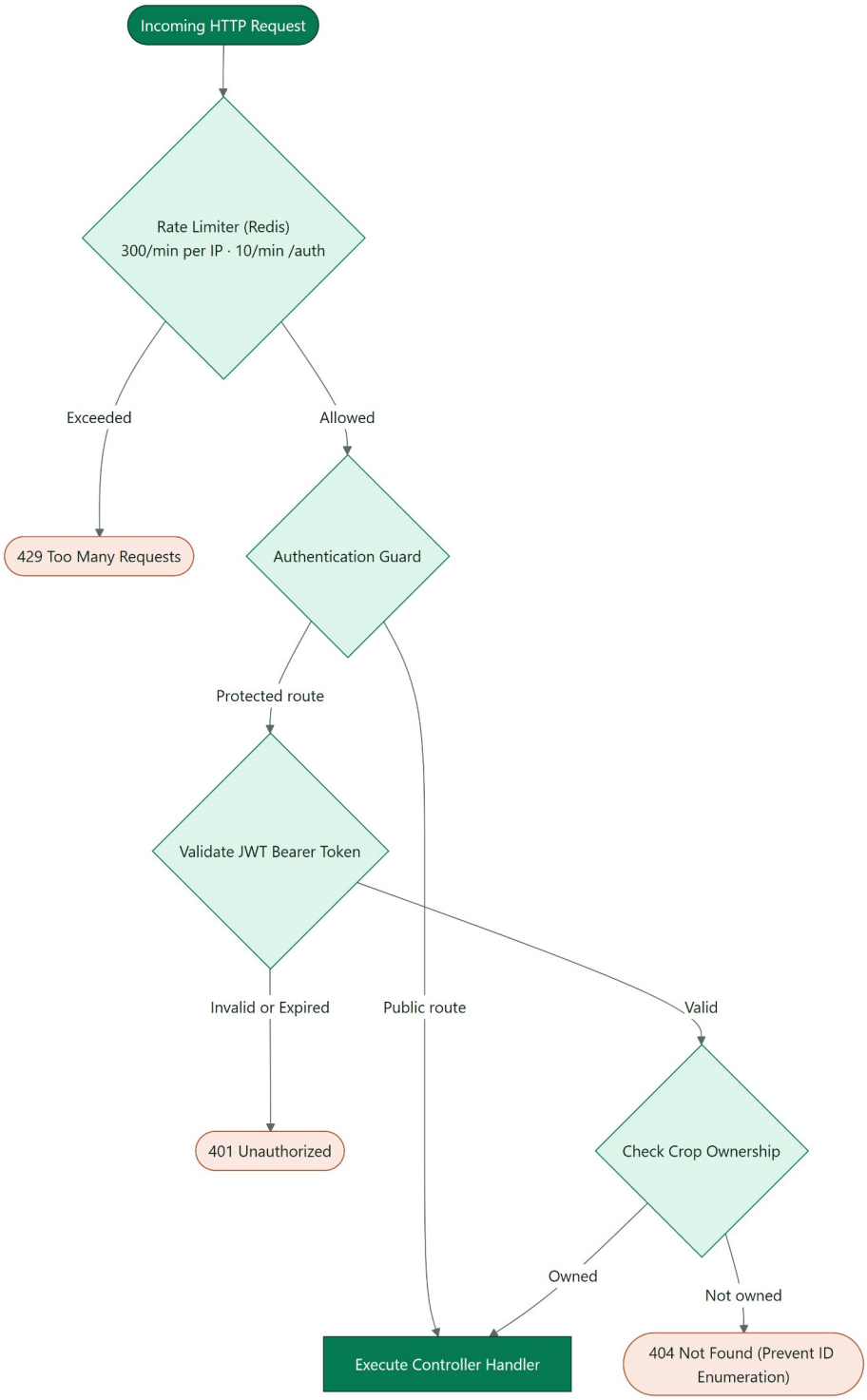


Figure / Figura 4 · *Request Security Validation Pipeline*

SECURITY CONTROL	IMPLEMENTATION DETAILS
Passwords	8 chars to 72 bytes max (upper, lower, digits); BCrypt cost factor 12
Password Recovery	Returns 202 Accepted on non-existent email; 30-min token stored as SHA-256
Device Credentials	24 cryptographically random bytes, encrypted with AES-256-GCM via `SMARTPOT_AES_KEY`
Internal Services	AI engine and data generator queried with bearer service token over docker network
Telegram Bot Integration	Webhook with constant-time token comparison; single-use 10-minute linking codes

SECTION 06

API and MQTT Contracts

Interactive OpenAPI documentation is hosted at `/docs` with raw JSON schema at `/v3/api-docs`. The repository README details REST endpoints, while the Platform Technical Documentation details the MQTT v1 protocol.

ROUTE GROUP	ENDPOINTS
Public	`/health`, `/api/v1/auth/*`, `/api/v1/crop-profiles`
Crops	`/api/v1/crops`, `/crops/{id}`, `/crops/automation`, `/crops/{id}/device`, `/crops/{id}/weather`
Crop Telemetry	`/readings`, `/actuators`, `/commands`, `/insights`, `/virtual-device`, `/channels`
Account & Fleet	`/api/v1/users/me`, `/notifications`, `/channels`, `/overview`, `/commands`, `/ai/learning`

PART III

Operations & Verification

SECTION 07

Configuration & Environment

The API loads `.env` on startup; all supported variables are declared in `.env.example`.

CONFIG CATEGORY	ENVIRONMENT VARIABLES
Server	`PORT`, `LOG_LEVEL`, `WEB_BASE_URL`, `PUBLIC_API_URL`, `CORS_ALLOWED_ORIGINS`, `API_DOCS_ENABLED`
Secrets	`SMARTPOT_JWT_SECRET` (32 caracteres o más), `SMARTPOT_AES_KEY` (AES-256 en Base64), `JWT_EXPIRATION`
Data Stores	`MONGODB_URI`, `REDIS_HOST`, `REDIS_PORT`, `REDIS_PASSWORD`
Mail Service	`MAIL_ENABLED`, `MAIL_HOST`, `MAIL_PORT`, `MAIL_USERNAME`, `MAIL_PASSWORD`, `MAIL_FROM`
MQTT	`MQTT_ENABLED`, `MQTT_BROKER_URI`, `MQTT_ADMIN_USERNAME`, `MQTT_ADMIN_PASSWORD`, `MQTT_PUBLIC_*`, `MQTT_WEBSOCKET_URL`
AI Engine	`AI_ENABLED`, `AI_BASE_URL`, `SMARTPOT_AI_TOKEN`, `AI_EVALUATION_INTERVAL`, `AI_AUTOMATION_COOLDOWN`, `AI_LEARNING_*`
Telegram	`TELEGRAM_BOT_TOKEN`, `TELEGRAM_BOT_USERNAME`, `TELEGRAM_MODE` (`polling` o `webhook`), `TELEGRAM_WEBHOOK_SECRET`
Simulator	`SIMULATOR_ENABLED`, `SIMULATOR_BASE_URL`, `SIMULATOR_TOKEN` (sin él no se crean cultivos virtuales)

AES KEY PERSISTENCE

If `SMARTPOT_AES_KEY` is lost or modified, stored MQTT device credentials cannot be decrypted: growers must rotate credentials for all physical crops.

SECTION 08

Automated Testing Suite

`./mvnw verify` executes 147 automated unit and integration tests via JUnit 5 and Mockito: cryptography, JWT validation, password constraints, MQTT v1 routing and telemetry parsing, broker provisioning, actuator switches, crop immutable typing, weather integrations, per-crop notifications, reactive agent

logic, continuous learning ingest, Telegram bot webhooks, simulation proxy, legacy backfill, Redis cache fallbacks, and security filters. The centralized E2E suite in `.github/docker/demo/test.py` validates 46 end-to-end assertions.

SECTION 09

Deployment and Production Operations

OPERATIONAL TASK	PROCEDURE
Health Check	`GET /health` → database, broker, cache, AI engine, and simulator (503 on database outage)
Log Inspection	`docker logs -f smartpot-api`
Container Image	`ghcr.io/smartpottech/smartpot-api`: multi-stage build, non-root user `1000`, read-only rootfs with `/tmp` tmpfs, and Docker `HEALTHCHECK`
CI/CD Pipeline	Every commit to `main` executes CI workflows, builds and signs container images on GHCR
Key Rotation	Device tab in PWA or `POST /api/v1/crops/{id}/device/key` (physical crops only)